Jaydeep Jitendra Borkar

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http://jaydeepborkar.github.io/
https://github.com/jaydeepborkar

Education & Research Experience

Northeastern University *Sept. 2021 - March 2026 (Expected)*
PhD in Computer Sciences
Advisor: Prof. David A. Smith

Meta Superintelligence Labs *June '26 - Oct. '26*
Research Scientist Intern (Safety Evals Team), New York
Research on LLM safety
Manager: Cyrus Nikolaidis

Meta Superintelligence Labs *June '25 - Nov '25*
Visiting Researcher (Pretraining Trust Team), New York
Research on LLM memorization & privacy
Manager: Archi Mitra
Mentors: Zheng Xu, Diego Garcia-Olano, Karan Chadha, Niloofar Mireshghallah

MIT-IBM Watson AI Lab *2020 - 2021*
External Research Student (remote)
Research on adversarial machine learning.
Advisor: Dr. Pin-Yu Chen

University of Pune *2016 - 2020*
Bachelor's degree in Computer Engineering

Research Interests: I'm broadly interested in studying safety in language models (e.g., training data extraction/memorization) and evaluating models for misalignment. I'm also interested in understanding the influence of pretraining data on various model capabilities and methods for curating high-quality synthetic datasets. My memorization research has been cited in a **U.S. congressional testimony** on AI privacy.

Papers & Research Projects

Memorization Dynamics in Knowledge Distillation for Language Models
Jaydeep Borkar, Karan Chadha, Niloofar Mireshghallah, Yuchen Zhang, Irina-Elina Veliche, Archi Mitra, David A. Smith, Zheng Xu, Diego Garcia-Olano
Conference on Language Modeling (COLM) 2026

Privacy Ripple Effects from Adding or Removing Personal Information in Language Model Training
Jaydeep Borkar, Matthew Jagielski, Katherine Lee, Niloofar Mireshghallah, David A. Smith, Christopher A. Choquette-Choo
Association for Computational Linguistics (ACL) 2025

Recite, Reconstruct, Recollect: Memorization in LMs as a Multifaceted Phenomenon
USVSN Sai Prashanth, Alvin Deng, Kyle O'Brien, Jyothir S V, Mohammad Aflah Khan, **Jaydeep Borkar**, Christopher A. Choquette-Choo, Jacob Ray Fuehne, Stella Biderman, Tracy Ke, Katherine Lee, Naomi Saphra
International Conference on Learning Representations (ICLR) 2025

What can we learn from Data Leakage and Unlearning for Law?
Jaydeep Borkar
Generative AI and Law (GenLaw) workshop, ICML 2023
https://genlaw.github.io/CameraReady/12.pdf

Mind the gap: Analyzing lacunae with transformer-based transcription
Jaydeep Borkar and David A. Smith

Workshop on Computational Paleography, ICDAR 2024
https://arxiv.org/abs/2407.00250

Simple Transparent Adversarial Examples
Jaydeep Borkar and Pin-Yu Chen
Workshop on Security and Safety in Machine Learning Systems, ICLR 2021
https://aisecure-workshop.github.io/aml-iclr2021/papers/48.pdf

Talks

Meta Mechanistic Interpretability Group: talk on memorization during distillation
Google ML Privacy Seminar: talk on memorization and privacy.
Northeastern CS 5100: invited guest lecture on privacy and security in LLMs.

Organizing

Trustworthy ML Initiative
Co-organizer of the Trustworthy ML Initiative along with Prof. Hima Lakkaraju (Harvard), Sara Hooker (Cohere for AI), Dr. Sarah Tan (Salesforce AI), Dr. Subho Majumdar (Vijil), Chhavi Yadav (UC San Diego), Dr. Chirag Agarwal (Harvard), Prof. Haohan Wang (UIUC), and Marta Lemanczyk (Hasso-Plattner-Institut).

Courses

Machine Learning CS 6140, Natural Language Processing CS 6120, Deep Learning CS 7150, Machine Learning Security and Privacy CY 7790, Theory and Methods in Human-Computer Interaction CS 7340, AI as an Archival Science CS 7180.

Teaching Experience

Natural Language Processing CS 6120 - TA	Summer 2024
Foundations of AI CS 5100- TA	Spring 2024
Foundations of Data Science DS 3000 - TA	Fall 2023
Product Development for Large Language Models CS 7180 - TA	Summer 2023
Introduction to Computer Science Research CS 3950 and CS 4950 - TA	Spring 2023
Introduction to Machine Learning and Data Mining DA 5030 - TA	Summer and Fall 2022